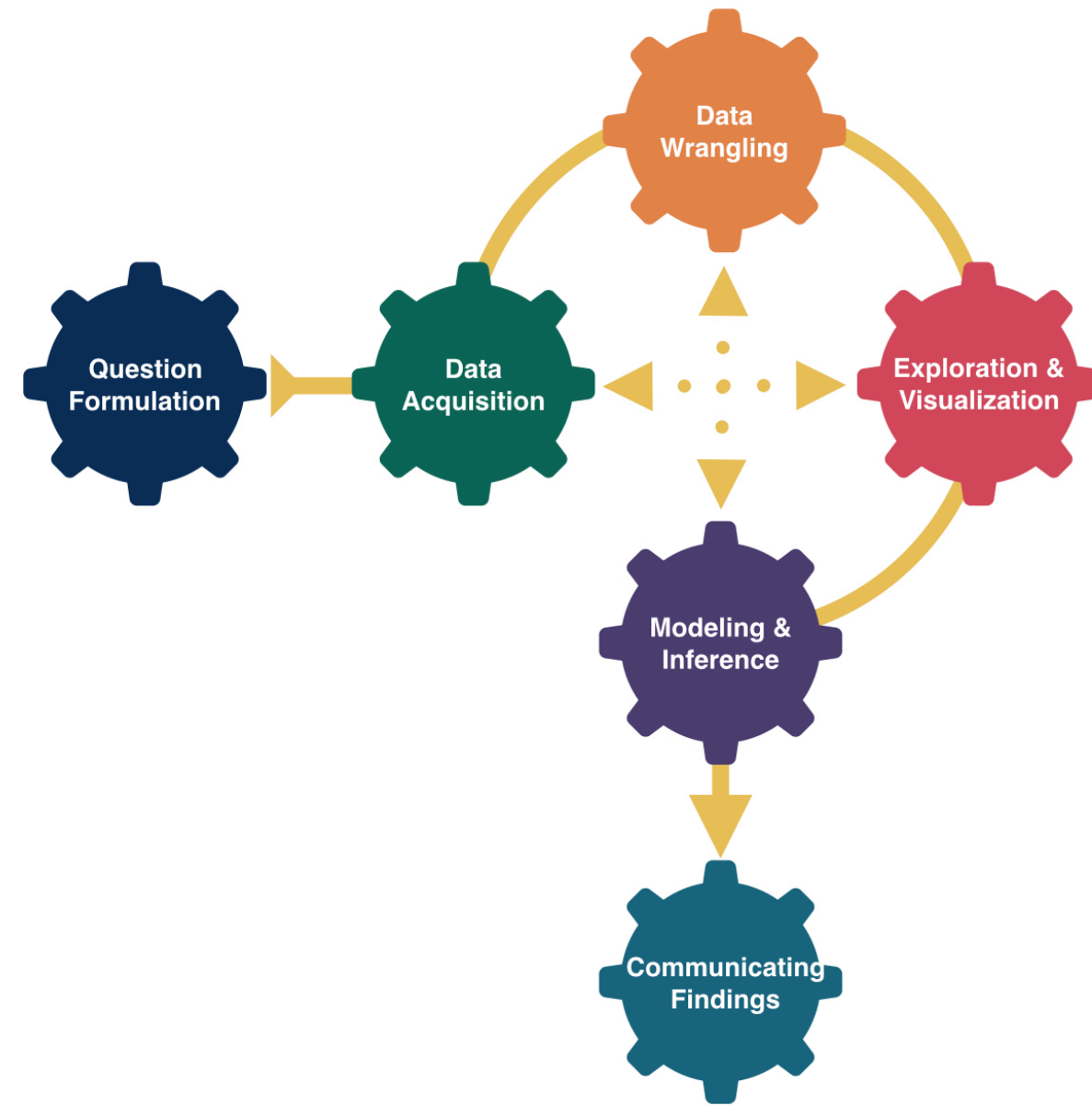


Other Models



Kelly McConville

DATA 100

Week 13 | Spring 2026

Announcements

- Extra Credit Lecture Quiz
- Don't forget to get your data science credit.

Goals for Today

- Other Models
- Analyzing categorical data

What other models/techniques to learn?

Two examples:

- Chi-squared tests:
 - Case where the test statistic is more involved.
- Logistic regression
 - Response variable: binary, categorical variable
 - Explanatory variables: Can be a mix of categorical and quantitative variables
- Non-parametric modeling
 - Don't require a specific global relationship between response variable and explanatory variable(s)

Inference for Categorical Variables

Consider the situation where:

- Response variable: categorical
- Explanatory variable: categorical
- Parameter of interest: $p_1 - p_2$
 - This parameter of interest only makes sense if **both** variables only have two categories.

It is time to learn how to study the relationship between two categorical variables when **at least one has more than two categories**.

Hypotheses

H_o : The two variables are independent.

H_a : The two variables are dependent.

Example

Near-sightedness typically develops during the childhood years. Quinn, Shin, Maguire, and Stone (1999) explored whether there is a relationship between the type of light children were exposed to and their eye health based on questionnaires filled out by the children's parents at a university pediatric ophthalmology clinic.

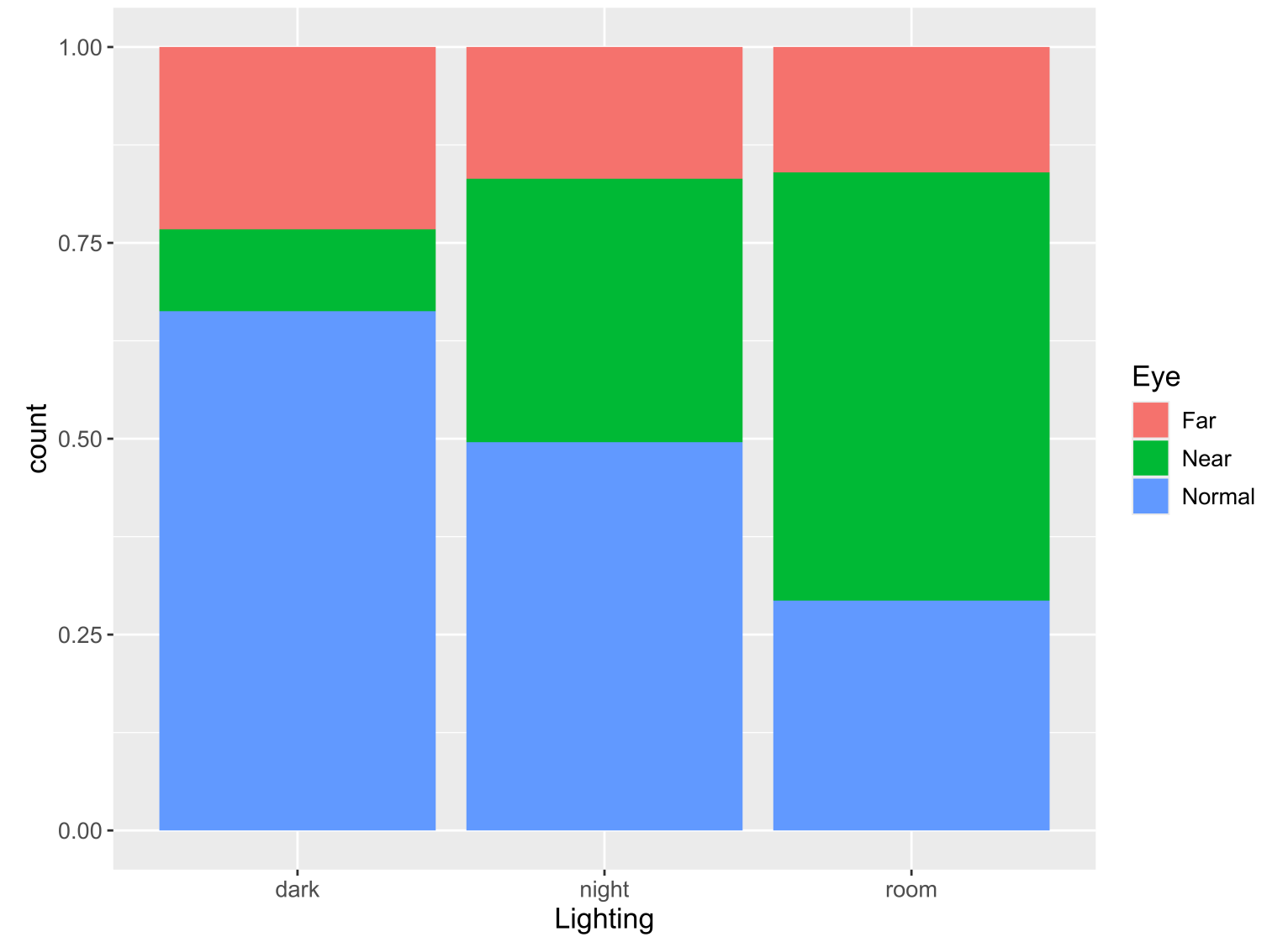
```
1 library(tidyverse)
2 library(infer)
3
4 # Import data
5 eye_data <- read_csv("data/eye_lighting.csv")
6
7 # Contingency table
8 eye_data %>%
9   count(Lighting, Eye)
```

```
# A tibble: 9 × 3
  Lighting Eye      n
  <chr>    <chr> <int>
1 dark    Far      40
2 dark    Near     18
3 dark    Normal  114
4 night   Far      39
5 night   Near     78
6 night   Normal  115
7 room    Far      12
8 room    Near     41
9 room    Normal   22
```

Eyesight Example

Does there appear to be a relationship/dependence?

```
1 ggplot(data = eye_data,  
2       mapping = aes(x = Lighting,  
3                     fill = Eye)) +  
4   geom_bar(position = "fill")
```



Test Statistic

Need a test statistic!

- Won't be a single sample statistic.
- Needs to measure the discrepancy between the observed sample and the sample we'd expect to see if H_o (no relationship) were true.
- Would be nice if its null distribution could be approximated by a known probability model.

Sample Result Tables

Observed Sample Table

```
1 table(eye_data$Eye, eye_data$Lighting) %>%
2   addmargins() %>%
3   kable(format = "html")
```

	dark	night	room	Sum
Far	40	39	12	91
Near	18	78	41	137
Normal	114	115	22	251
Sum	172	232	75	479

Expected Sample Table

Question: If H_o were correct, is this the table that we'd expect to see?

	dark	night	room	Sum
Far	53	53	53	159
Near	53	53	53	159
Normal	53	53	53	159
Sum	159	159	159	477

Sample Result Tables

Observed Sample Table

```
1 table(eye_data$Eye, eye_data$Lighting) %>%  
2   addmargins() %>%  
3   kable(format = "html")
```

	dark	night	room	Sum
Far	40	39	12	91
Near	18	78	41	137
Normal	114	115	22	251
Sum	172	232	75	479

Expected Sample Table

Question: If H_o were correct, what table would we expect to see?

Want a H_o table that respects the overall eye condition proportions:

$$\hat{p}_{far} = 91/479$$

$$\hat{p}_{nor} = 251/479$$

$$\hat{p}_{nea} = 137/479$$

Sample Result Tables

Observed Sample Table

```
1 table(eye_data$Eye, eye_data$Lighting) %>%
2   addmargins() %>%
3   kable(format = "html")
```

	dark	night	room	Sum
Far	40	39	12	91
Near	18	78	41	137
Normal	114	115	22	251
Sum	172	232	75	479

Expected Sample Table

Question: If H_o were correct, what table would we expect to see?

	dark	night	room	Sum
Far	(91/479)172	(91/479)232	(91/479)75	91
Near	(137/479)172	(137/479)232	(137/479)75	137
Normal	(251/479)172	(251/479)232	(251/479)75	251
Sum	172	232	75	479

Still have the same totals but distributed the values differently within the table

Sample Result Tables

Observed Sample Table

```
1 table(eye_data$Eye, eye_data$Lighting) %>%
2   addmargins() %>%
3   kable(format = "html")
```

	dark	night	room	Sum
Far	40	39	12	91
Near	18	78	41	137
Normal	114	115	22	251
Sum	172	232	75	479

Expected Sample Table

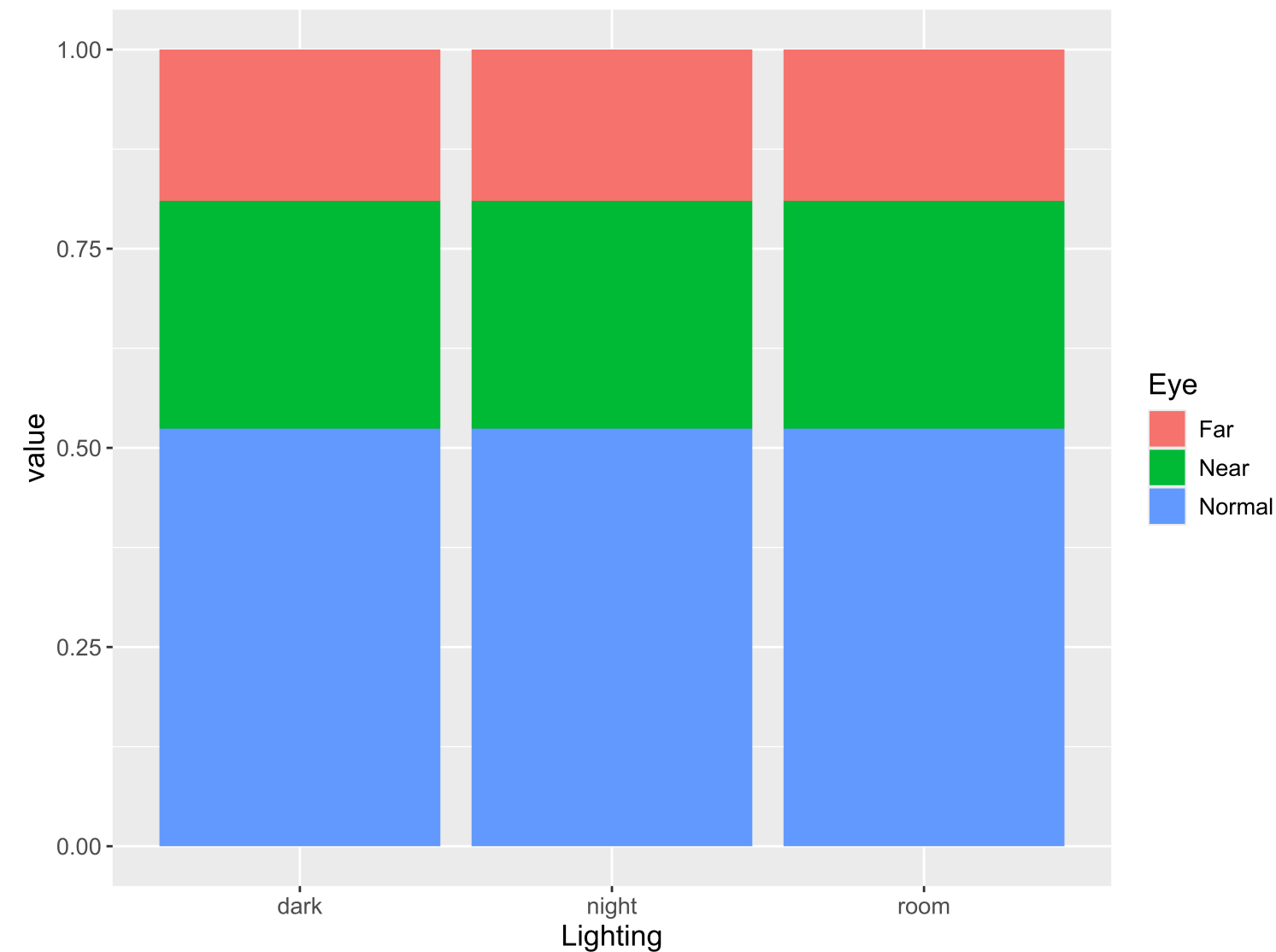
Question: If H_o were correct, what table would we expect to see?

	dark	night	room	Sum
Far	32.7	44.1	14.2	91.0
Near	49.2	66.4	21.5	137.1
Normal	90.1	121.6	39.3	251.0
Sum	172.0	232.1	75.0	479.1

Expected Table

How does this table represent H_o ?

	dark	night	room	Sum
Far	32.68	44.08	14.25	91.01
Near	49.19	66.35	21.45	136.99
Normal	90.13	121.57	39.30	251.00
Sum	172.00	232.00	75.00	479.00



Test Statistic

Want the test statistic to quantify the difference between the observed table and the expected table.

	dark	night	room	Sum
Far	40	39	12	91
Near	18	78	41	137
Normal	114	115	22	251
Sum	172	232	75	479

	dark	night	room	Sum
Far	32.68	44.08	14.25	91
Near	49.19	66.35	21.45	137
Normal	90.13	121.57	39.30	251
Sum	172.00	232.00	75.00	479

For each cell: Compute a **Z-score**!

$$\begin{aligned} \text{Z-score} &= \frac{\text{stat} - \text{mean}}{\text{SE}} \\ &= \frac{\text{observed} - \text{expected}}{\sqrt{\text{expected}}} \end{aligned}$$

Test Statistic

Want the test statistic to quantify the difference between the observed table and the expected table.

	dark	night	room
Far	1.28	-0.76	-0.60
Near	-4.45	1.43	4.22
Normal	2.51	-0.60	-2.76

Test Statistic

Test Statistic Formula:

$$\chi^2 = \sum \left(\frac{\text{observed} - \text{expected}}{\sqrt{\text{expected}}} \right)^2$$

```
1 library(infer)
2 #Compute Chi-square test stat
3 test_stat <- eye_data %>%
4   specify(Eye ~ Lighting) %>%
5   calculate(stat = "Chisq")
6 test_stat
```

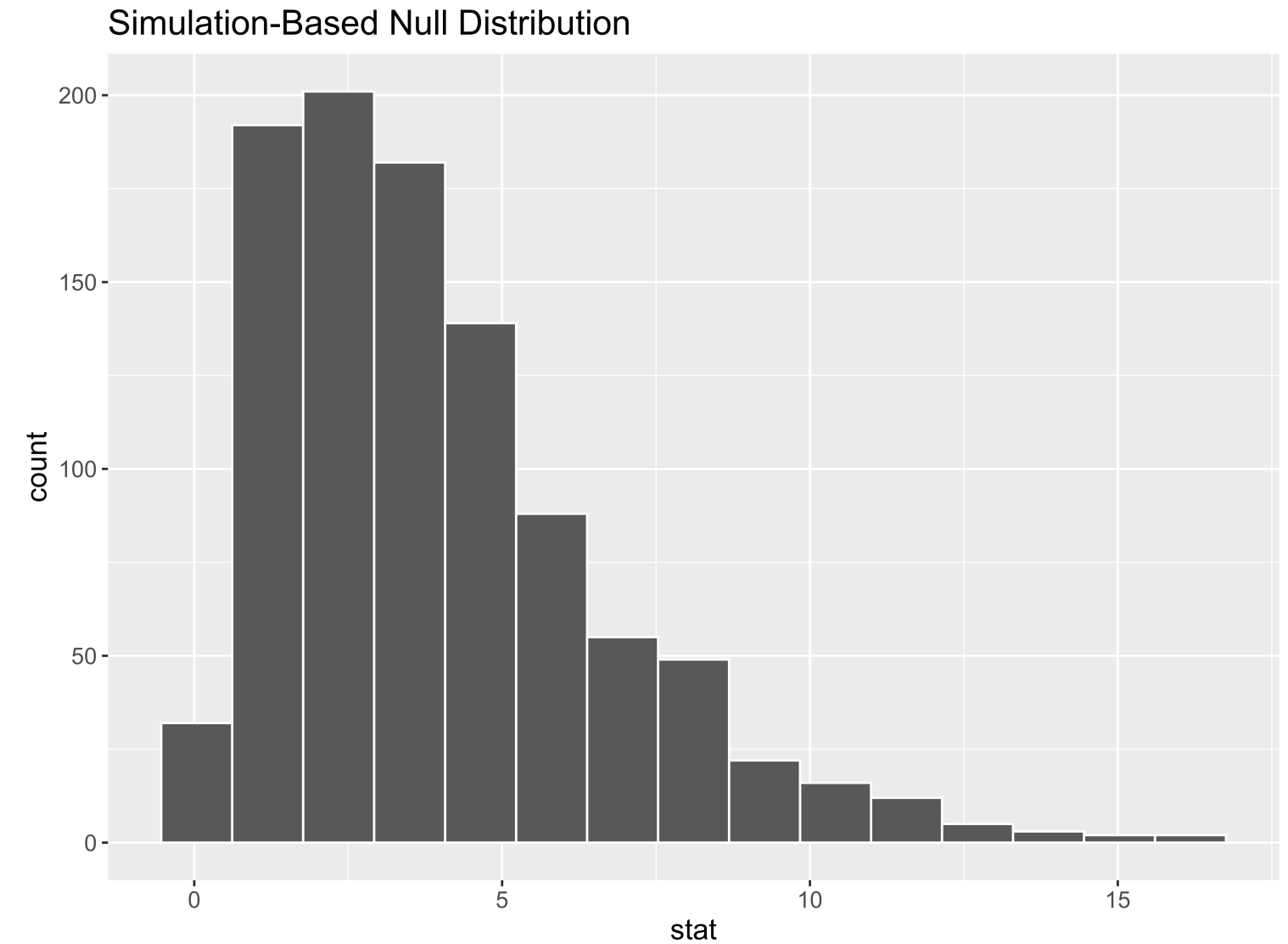
```
Response: Eye (factor)
Explanatory: Lighting (factor)
# A tibble: 1 × 1
  stat
  <dbl>
1  56.5
```

Questions:

1. Is a test statistic **unusual** if it is a large number or a small number?
2. Is 56.5 **unusual** under H_o ?

Generating the Null Distribution

```
1 # Construct null distribution
2 null_dist <- eye_data %>%
3   specify(Eye ~ Lighting) %>%
4   hypothesize(null = "independence") %>%
5   generate(reps = 1000, type = "permute") %>%
6   calculate(stat = "Chisq")
7
8
9 visualize(null_dist)
```



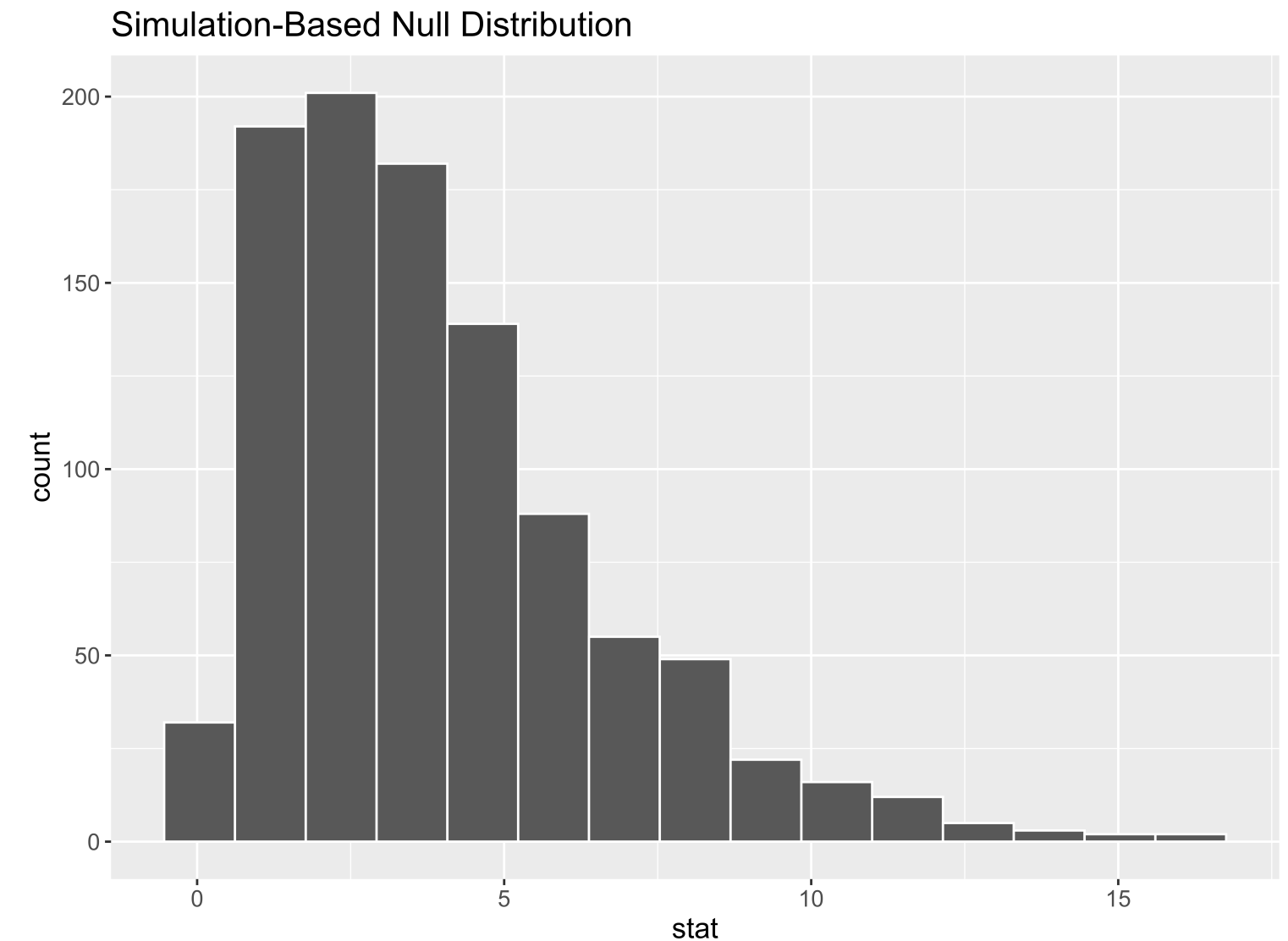
The Null Distribution

Key Observations about the distribution:

Smallest possible value?

Shape?

Is our observed test statistic of 56.5 unusual?



The P-value

```
1 # Compute p-value
2 null_dist %>%
3   get_pvalue(obs_stat = test_stat, direction = "greater")

# A tibble: 1 × 1
  p_value
  <dbl>
1       0
```

Approximating the Null Distribution

If there are at least 5 observations in each cell, then the test statistic can be approximated by a

$$\chi^2(df = (k - 1)(j - 1)) \text{ random variable}$$

where

- k is the number of categories in the response variable
- j is the number of categories in the explanatory variable.
- The df controls the center and spread of the distribution.

The Chi-Squared Test

```
1 chisq_test(eye_data, Eye ~ Lighting)
```

```
# A tibble: 1 × 3  
  statistic chisq_df p_value  
    <dbl>    <int>   <dbl>  
1     56.5        4 1.56e-11
```

- Conclusions?
- Causal link between room lighting at bedtime and eye conditions?
- Decisions, decisions...

Another Example

In your opinion, which sentence is more grammatically correct?

Version 1: “It’s important for a person to be honest, kind and loyal.”

Version 2: “It’s important for a person to be honest, kind, and loyal.”

How would you write the following sentence?

Version 1: “Some experts say it’s important to drink milk, but the data are inconclusive.”

Version 2: “Some experts say it’s important to drink milk, but the data is inconclusive.”

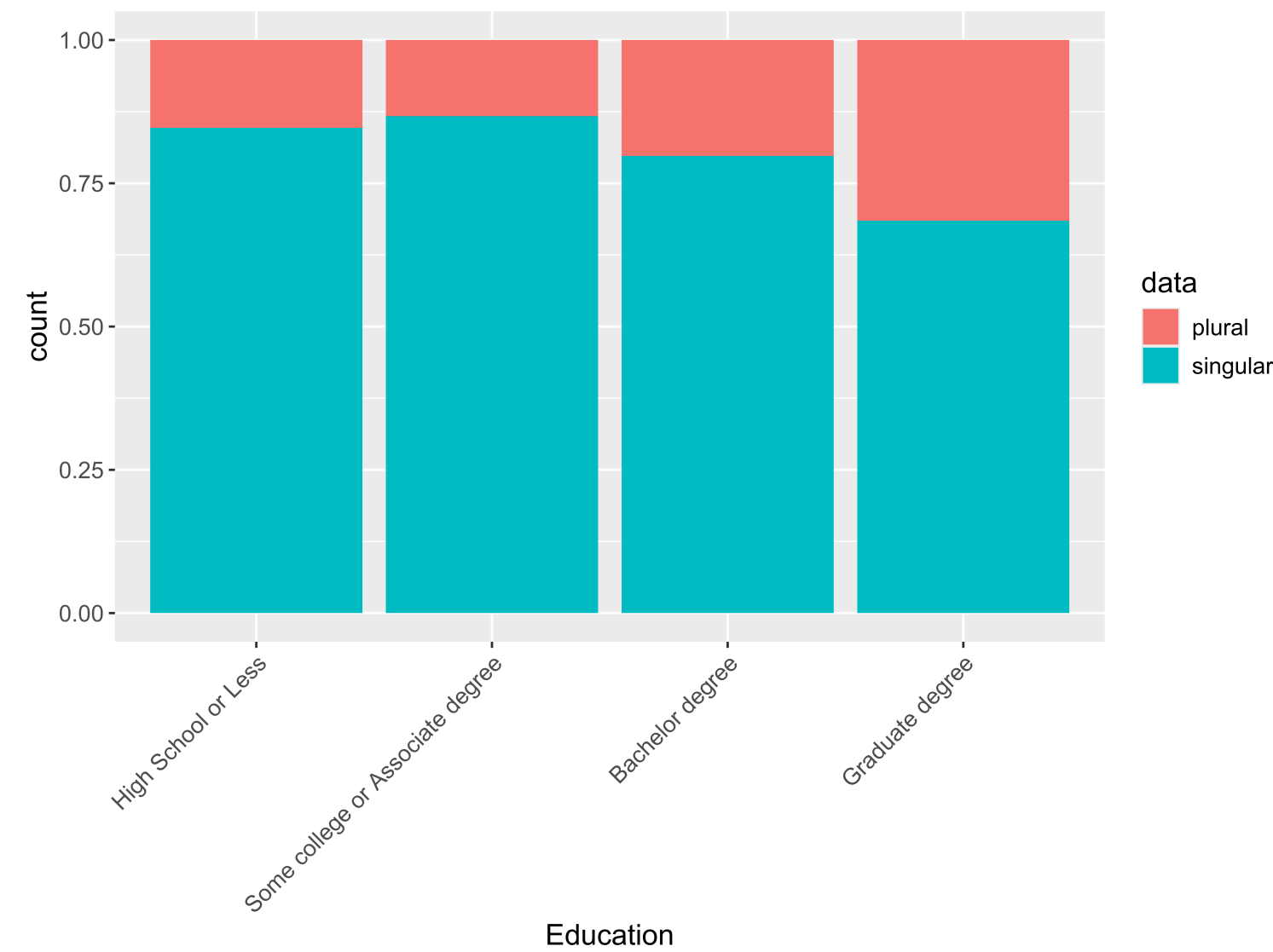
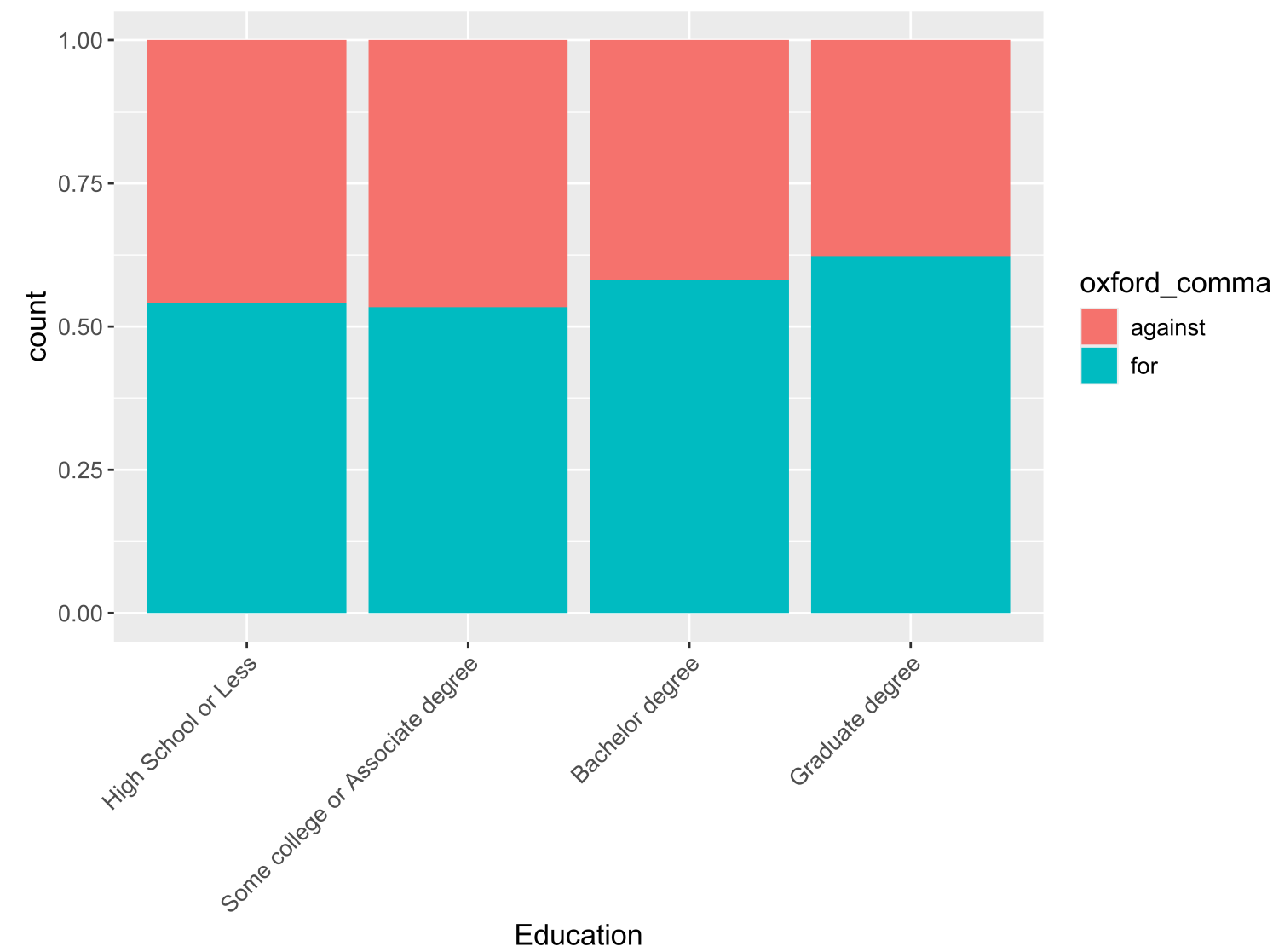
Grammar and Education

In 2014, [Fivethirtyeight.com](https://fivethirtyeight.com) and [SurveyMonkey](https://surveymonkey.com) ran a poll asking people about their opinions on the oxford comma and data. We want to use these data to investigate if there is a relationship between someone's level of education and their grammar opinions.

```
1 # Data loading and wrangling
2
3 grammar <- read_csv("https://raw.githubusercontent.com/fivethirtyeight/data/master/comma-survey/comma-survey.csv") %>
4   mutate(oxford_comma = if_else(`In your opinion, which sentence is more gramatically correct?` == "It's important fo
5                                   "for",
6                                   "against")) %>%
7   mutate(Education = fct_collapse(Education,
8                                   `High School or Less` = c("Less than high school degree",
9                                                             "High school degree"))) %>%
10  mutate(Education = fct_relevel(Education, "High School or Less", "Some college or Associate degree")) %>%
11  mutate(data = if_else(`How would you write the following sentence?` == "Some experts say it's important to drink mi
12  drop_na(data, Education, oxford_comma)
```

Grammar and Education

What story do these graphs tell? Does education level appear to have a stronger relationship with preference for the oxford comma or with whether the word “data” is singular or plural?



Grammar and Education

Check conditions!

```
1 count(grammar, Education, oxford_comma)
```

# A tibble: 8 × 3		
Education	oxford_comma	n
<fct>	<chr>	<int>
1 High School or Less	against	51
2 High School or Less	for	60
3 Some college or Associate degree	against	137
4 Some college or Associate degree	for	157
5 Bachelor degree	against	143
6 Bachelor degree	for	198
7 Graduate degree	against	104
8 Graduate degree	for	172

```
1 count(grammar, Education, data)
```

# A tibble: 8 × 3		
Education	data	n
<fct>	<chr>	<int>
1 High School or Less	plural	17
2 High School or Less	singular	94
3 Some college or Associate degree	plural	39
4 Some college or Associate degree	singular	255
5 Bachelor degree	plural	69
6 Bachelor degree	singular	272
7 Graduate degree	plural	87
8 Graduate degree	singular	189

Grammar and Education

Run tests.

```
1 chisq_test(grammar, oxford_comma ~ Education)
```

```
# A tibble: 1 × 3  
  statistic chisq_df p_value  
    <dbl>    <int>   <dbl>  
1      5.22        3  0.156
```

```
1 chisq_test(grammar, data ~ Education)
```

```
# A tibble: 1 × 3  
  statistic chisq_df      p_value  
    <dbl>    <int>    <dbl>  
1      31.5        3 0.000000652
```

- Why do both tests have 3 degrees of freedom?
- Which test statistic is more unusual under the null? Does this agree with what you saw in the graph?
- How to interpret one of the p-values in context?
- Conclusions?

Other Models

Logistic Regression

Response variable: A **categorical** variable with **two** categories

$$Y = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

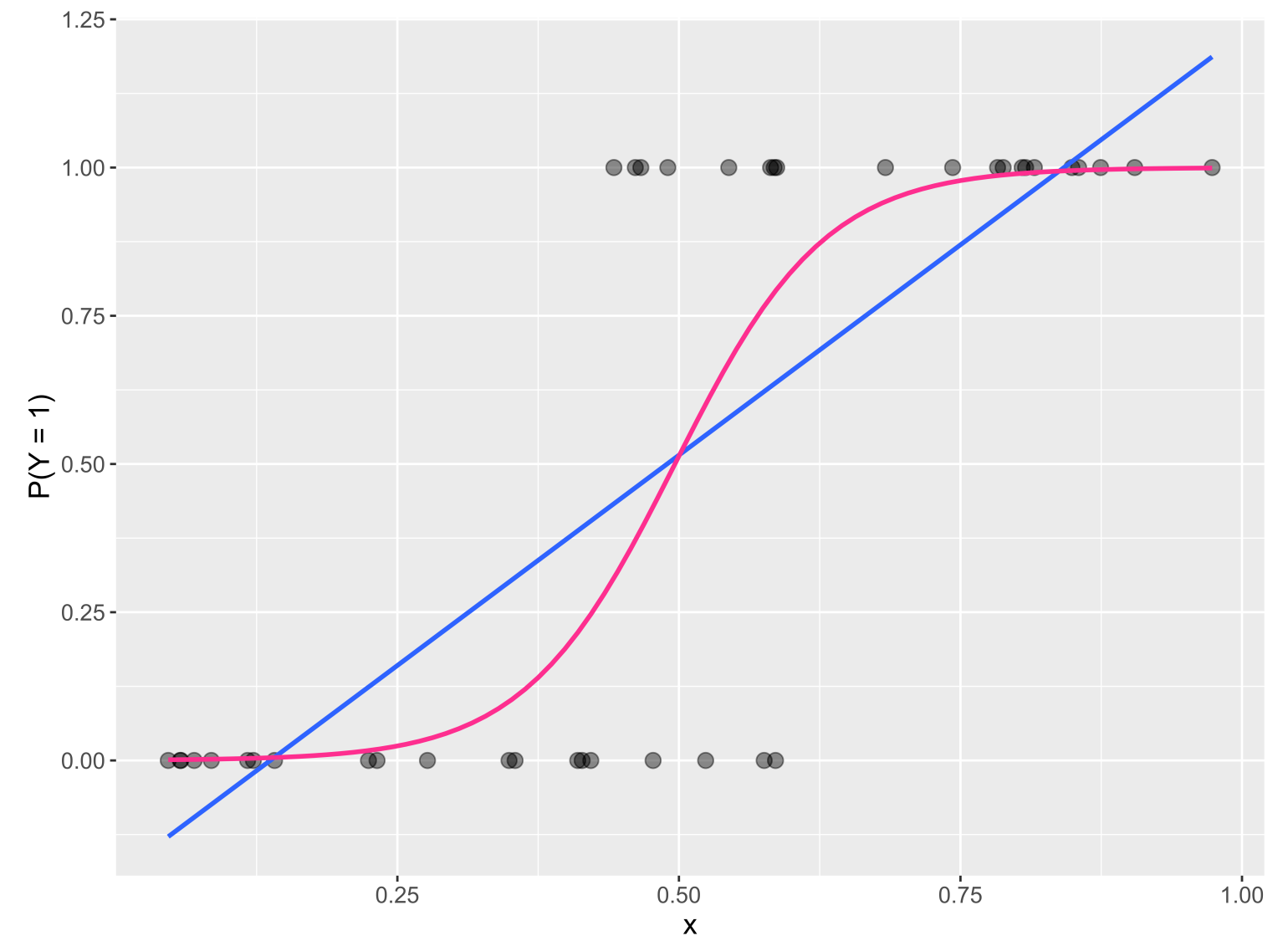
$Y \sim \text{Bernoulli}(p)$ where $p = P(Y = 1) = P(\text{success})$.

Explanatory variables: Can be a mix of categorical and quantitative

Goal: Build a model for $P(Y = 1)$.

Why can't we use linear regression?

- Regression line = estimated **probability** of success
- For valid values of x , we are predicting the probability is less than 0 or greater than 1.
- The estimated probabilities based on the logistic regression model are bounded between 0 and 1.



What does the **logistic regression model** look like?

Logistic Regression Model

$$\log \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) = \beta_o + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

Left hand side has many interpretations:

$$\begin{aligned} \text{LHS} &= \log \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) \\ &= \log (\text{odds (of success)}) \\ &= \text{logit}(P(Y = 1)) \end{aligned}$$

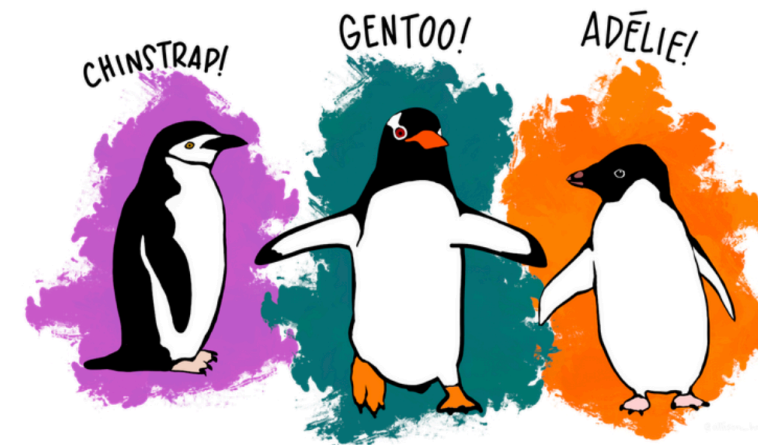
Note:

$$\text{odds} = \frac{\text{prob of success}}{\text{prob of failure}}$$

Example

Let's return to the Palmer Penguins dataset and build a model that predicts whether a penguin is a **Chinstrap** or a **Adelie**.

```
1 knitr::include_graphics("img/penguins.png")
```



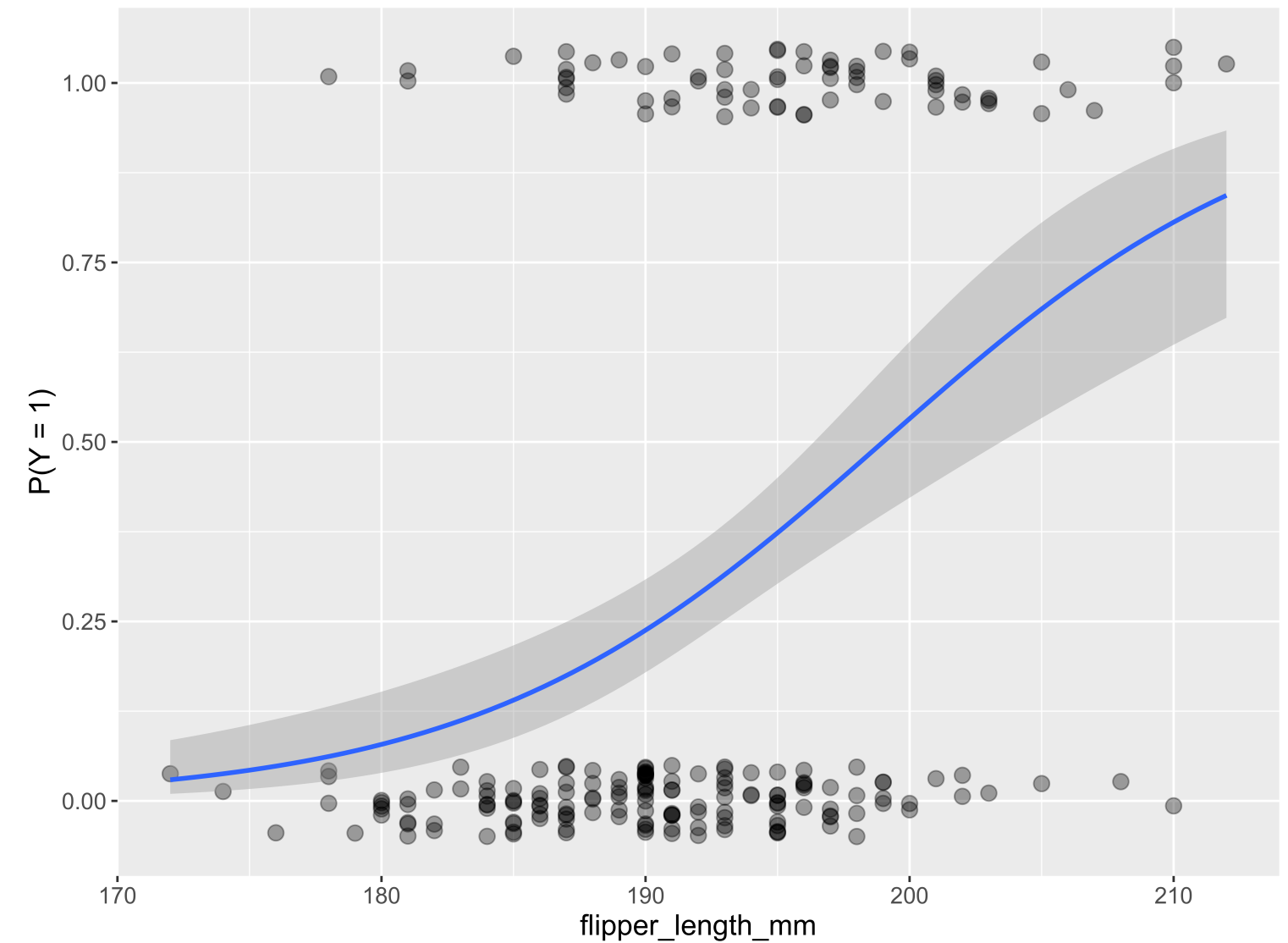
The Palmer Archipelago penguins. Artwork by @allison_horst.

```
1 library(palmerpenguins)
2
3 penguins2 <- penguins %>%
4   filter(species != "Gentoo") %>%
5   mutate(species_num =
6     if_else(species == "Chinstrap",
7             1, 0)) %>%
8   drop_na(species_num, flipper_length_mm)
9
10 count(penguins2, species)
```

```
# A tibble: 2 × 2
  species      n
```

Example

```
1 ggplot(data = penguins2,  
2       mapping = aes(x = flipper_length_mm,  
3                     y = species_num)) +  
4   geom_jitter(size = 3, alpha = 0.4,  
5              width = 0, height = 0.05) +  
6   labs(y = "P(Y = 1)") +  
7   geom_smooth(method = 'glm',  
8              method.args =  
9               list(family = "binomial"))
```



Example

```
1 mod <- glm(species ~ flipper_length_mm, family = "binomial",  
2           data = penguins2)  
3 library(broom)  
4 tidy(mod)
```

A tibble: 2 × 5

	term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
1	(Intercept)	-25.8	4.90	-5.26	0.000000141
2	flipper_length_mm	0.129	0.0253	5.12	0.000000299

But what do the coefficient estimates $\hat{\beta}_0 = -25.8$ and $\hat{\beta}_1 = 0.129$ represent?

Interpreting Coefficients

$$\exp(\hat{\beta}_1) = \frac{\text{odds of being Chinstrap for flipper length } t + 1 \text{ mm}}{\text{odds of being Chinstrap for flipper length } t \text{ mm}}$$

```
1 exp(0.129)
```

```
[1] 1.13769
```

The odds that a penguin in Chinstrap instead of Adelie is 1.138 times the odds of a penguin whose flipper is 1 mm shorter.

Inference

Is flipper length a useful explanatory variable?

```
1 tidy(mod)
```

```
# A tibble: 2 × 5
```

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	-25.8	4.90	-5.26	0.000000141
2	flipper_length_mm	0.129	0.0253	5.12	0.000000299

There are LOTS of models out there beyond regression! Next time, consider classification trees.

